CHAPTER 5:
Science and Medicine

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ACCESS THE PUBLIC DATA

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Overview

This chapter explores key trends in AI-driven science and medicine, reflecting the technology's growing impact in these fields. It begins with notable AI milestones from 2024, followed by an analysis of AI in protein folding, an important area of scientific advancement. The chapter then examines AI's role in clinical care, spanning both imaging and non-imaging applications. This includes a review of clinical knowledge capabilities in new language models, diagnostic and clinical management capabilities of AI systems, real-world AI deployments in medicine, synthetic data applications, and social determinants of health. Finally, the chapter concludes with an exploration of ethical trends in AI medical research.

This chapter was prepared by RAISE Health (Responsible AI for Safe and Equitable Health), a collaboration between Stanford Medicine and the Stanford Institute for Human-Centered Artificial Intelligence (HAI). Since its launch in 2023, RAISE Health has worked to advance responsible AI innovation in biomedical research, education, and patient care, with a focus on ensuring that these technologies benefit everyone.

Fostering collaborative research and knowledge sharing are central to RAISE Health's mission. As part of that commitment, RAISE Health partnered with the AI Index Steering Committee to expand the group's focus to include key developments in science and medicine. In 2024, this collaboration produced the inaugural chapter on science and medicine, highlighting major AI advancements at Stanford and beyond. The 2025 chapter builds on that foundation with contributions from members of the RAISE Health faculty research council, Stanford School of Medicine faculty, postdoctoral fellows, and undergraduate students from the schools of Medicine and Engineering.

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Chapter Highlights

1. Bigger and better protein sequencing models emerge. In 2024, several large-scale, high-performance protein sequencing models, including ESM3 and AlphaFold 3, were launched. Over time, these models have grown significantly in size, leading to continuous improvements in protein prediction accuracy.

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2. AI continues to drive rapid advances in scientific discovery. AI's role in scientific progress continues to expand. While 2022 and 2023 marked the early stages of AI-driven breakthroughs, 2024 brought even greater advancements, including Aviary, which trains LLM agents for biological tasks, and FireSat, which significantly enhances wildfire prediction.

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3. The clinical knowledge of leading LLMs continues to improve. OpenAI's recently released o1 set a new state-of-the-art 96.0% on the MedQA benchmark—a 5.8 percentage point gain over the best score posted in 2023. Since late 2022, performance has improved 28.4 percentage points. MedQA, a key benchmark for assessing clinical knowledge, may be approaching saturation, signaling the need for more challenging evaluations.

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4. AI outperforms doctors on key clinical tasks. A new study found that GPT-4 alone outperformed doctors—both with and without AI—in diagnosing complex clinical cases. Other recent studies show AI surpassing doctors in cancer detection and identifying high-mortality-risk patients. However, some early research suggests that AI-doctor collaboration yields the best results, making it a fruitful area of further research.

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5. The number of FDA-approved, AI-enabled medical devices skyrockets. The FDA authorized its first AI-enabled medical device in 1995. By 2015, only six such devices had been approved, but the number spiked to 223 by 2023.

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6. Synthetic data shows significant promise in medicine. Studies released in 2024 suggest that AI-generated synthetic data can help models better identify social determinants of health, enhance privacy-preserving clinical risk prediction, and facilitate the discovery of new drug compounds.

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Chapter Highlights (cont'd)

7. Medical AI ethics publications are increasing year over year. The number of publications on ethics in medical AI quadrupled from 2020 to 2024, rising from 288 in 2020 to 1,031 in 2024.

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8. Foundation models come to medicine. In 2024, a wave of large-scale medical foundation models were released, ranging from general-purpose multimodal models like Med-Gemini to specialized models such as EchoCLIP for echocardiology and ChexAgent for radiology.

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9. Publicly available protein databases grow in size. Since 2021, the number of entries in major public protein science databases has grown significantly, including UniProt (31%), PDB (23%), and AlphaFold (585%). This expansion has important implications for scientific discovery.

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10. AI research wins two Nobel Prizes. In 2024, AI-driven research received top honors, with two Nobel Prizes awarded for AI-related breakthroughs. Google DeepMind's Demis Hassabis and John Jumper won the Nobel Prize in Chemistry for their pioneering work on protein folding with AlphaFold. Meanwhile, John Hopfield and Geoffrey Hinton received the Nobel Prize in Physics for their foundational contributions to neural networks.

This section highlights significant AI-related medical and biological breakthroughs in 2024 as chosen by the RAISE Health AI Index Workgroup and AI Index Steering Committee.

5.1 Notable Medical and Biological AI Milestones

Protein Sequence Optimization

LLMs optimize protein sequence optimization

LLMs have recently, albeit unintentionally, gained a new biological capability: optimizing protein sequences. Traditionally, protein engineering requires extensive lab studies to refine sequences for improved functionality. However, a recent study found that LLMs—without fine-tuning—are becoming remarkably effective at this task. In other words, this is a hidden strength of existing LLMs, exemplified in this case by an adapted version of Llama-3.1-8B-Instruct. Using a directed evolutionary approach, researchers demonstrated that LLMs can generate protein sequences that outperform conventional algorithms across both synthetic and experimental fitness landscapes.

Figure 5.1.1 illustrates the researchers' findings. The objective in this case is to maximize the fitness value, with higher scores indicating better performance. The researchers compared their proposed method's fitness score against that of the default evolutionary algorithm (EA) approach.[1] The study revealed that this optimization extends beyond single-objective tasks to include constrained, budget-limited, and multiobjective scenarios. This compelling finding highlights the emergent properties of state-of-the-art LLMs, suggesting that as these general-purpose models continue to improve, their impact on scientific fields will only grow.

Single-objective optimization results for fitness optimization

Source: Wang et al., 2024

Dataset	Method	Population × iteration	Fitness score		
			Top 1	Top 10	Top 50
GB1	EA	32×4	**5.38±1.77**	**3.81±1.10**	**2.31±0.71**
		48×4	**4.88±0.33**	3.72±0.38	2.17±0.27
		96×4	**5.72±0.56**	**4.32±0.53**	2.84±0.60
	Ours	32×4	4.34±0.53	3.22±0.23	1.94±0.28
		48×4	4.31±0.82	**3.76±0.82**	**2.45±0.61**
		96×4	4.80±0.52	4.09±0.19	**3.04±0.19**
TrpB	EA	32×4	0.20±0.18	0.14±0.12	0.07±0.05
		48×4	0.67±0.14	0.52±0.11	0.19±0.04
		96×4	0.74±0.01	0.59±0.03	0.35±0.10
	Ours	32×4	**0.60±0.10**	**0.50±0.07**	**0.35±0.07**
		48×4	**0.68±0.04**	**0.58±0.01**	**0.36±0.01**
		96×4	**0.78±0.20**	**0.60±0.16**	**0.39±0.16**
Syn-3bfo	EA	32×8	0.57±0.21	-0.44±0.11	-1.35±0.17
		48×8	1.29±0.36	0.42±0.24	-0.63±0.07
		96×8	1.85±0.47	1.10±0.28	0.07±0.28
	Ours	32×8	**2.51±0.23**	**1.33±0.14**	**0.28±0.20**
		48×8	**2.35±0.26**	**1.36±0.11**	**0.04±0.09**
		96×8	**2.83±0.20**	**2.02±0.36**	**0.96±0.36**
AAV	EA	32×8	0.42±0.03	0.36±0.01	0.32±0.00
		48×8	0.44±0.00	0.38±0.01	0.33±0.00
		96×8	0.44±0.00	0.40±0.01	0.36±0.00
	Ours	32×8	**0.74±0.00**	**0.69±0.02**	**0.62±0.03**
		48×8	**0.75±0.01**	**0.71±0.01**	**0.64±0.02**
		96×8	**0.76±0.03**	**0.73±0.03**	**0.68±0.03**
GFP	EA	32×8	0.43±0.13	0.21±0.02	0.12±0.01
		48×8	0.43±0.14	0.26±0.05	0.12±0.01
		96×8	0.50±0.11	0.34±0.05	0.18±0.01
	Ours	32×8	**0.96±0.02**	**0.94±0.01**	**0.88±0.03**
		48×8	**0.96±0.02**	**0.93±0.01**	**0.84±0.02**
		96×8	**0.97±0.01**	**0.95±0.01**	**0.92±0.01**

Figure 5.1.1

1 Evolutionary algorithms (EA) simulate key aspects of biological evolution within a computer program to tackle complex problems—especially those without precise or fully satisfactory solutions—by finding approximate answers.

Aviary

Training LLM agents for biological tasks

As AI systems become increasingly useful, particularly for scientific use cases, one challenge has been designing language models that can interact with tools as they reason through complex tasks. Aviary introduces a structured framework for training language agents for three particularly challenging scientific tasks: DNA manipulation (for molecular cloning), answering research questions (through accessing scientific papers), and engineering protein stability. Figure 5.1.2 compares the performance of different models across various Aviary environments. It contrasts a baseline Claude 3.5 Sonnet model, which attempts tasks without environmental access, with models integrated into agent frameworks within the Aviary environment. Across nearly all tasks, the agentic models outperform the baseline. This research demonstrates that (1) although general-purpose LLMs perform well at many scientific tasks, fine-tuning models alongside domain experts often helps models yield superior results, and (2) AI-driven scientific research can be accelerated not only by model size but also through interaction with external tools, capabilities now commonly referred to as "agentic AI."

Performance of LLMs and language agents to solve tasks using Aviary environments
Source: Narayanan et al., 2024 | Chart: 2025 AI Index report

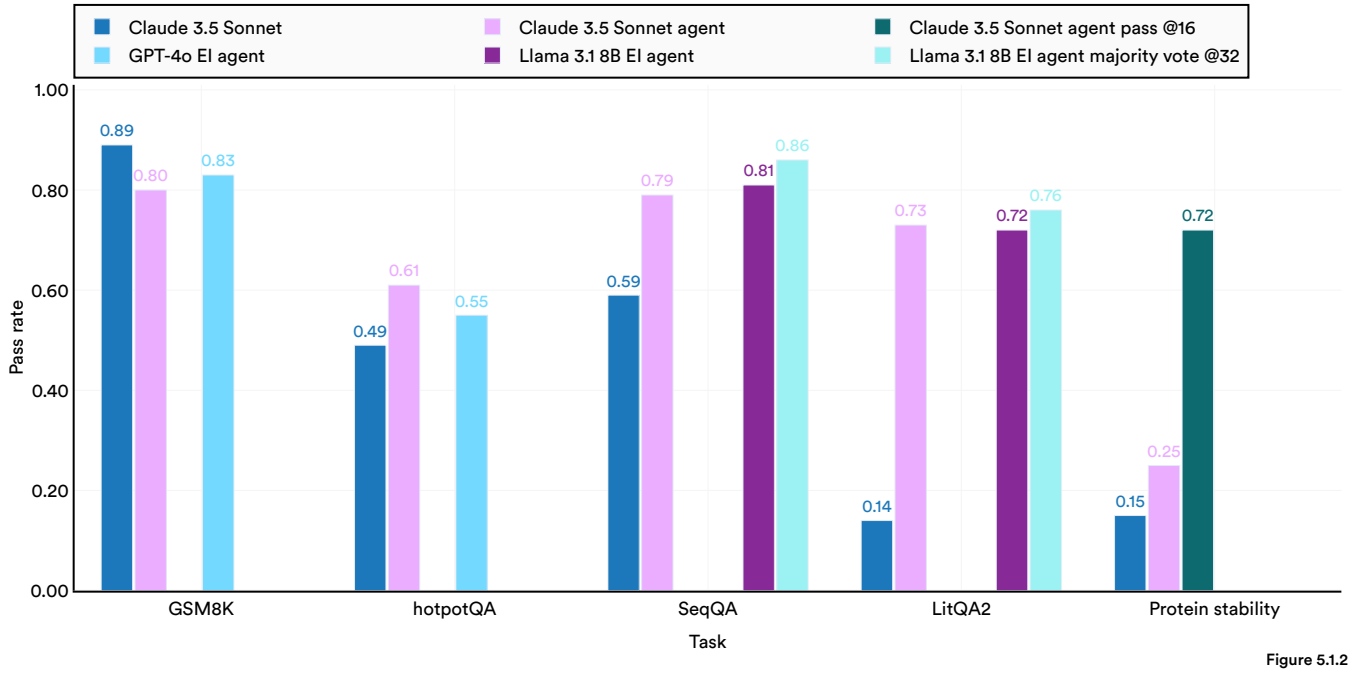


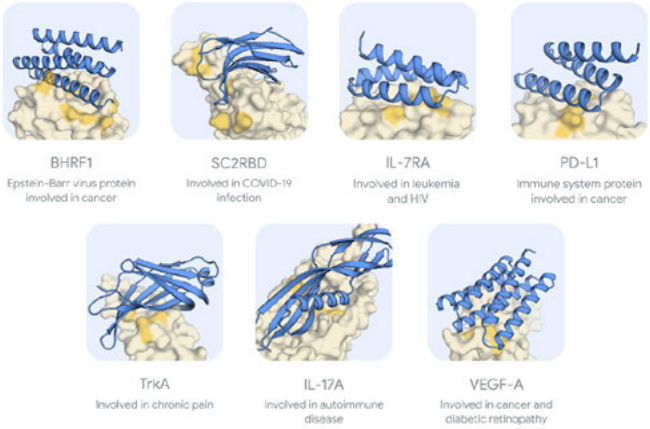
Figure 5.1.2

AlphaProteo

AI for novel, high-affinity protein binders

AlphaProteo is Google DeepMind's model focused on creating novel, high-affinity protein binders that attach to specific target molecules. Figure 5.1.3 illustrates the predicted structures of seven target proteins for which AlphaProteo created successful binders. AlphaProteo has designed the first protein binders for many targets, including VEGF-A, a protein linked to cancer and diabetes. Many of the tool's binding strengths are significantly better than current state-of-the-art solutions; in fact, the team estimates that some of their binders are up to 300 times more effective than anything currently available on the seven target proteins they tested. For the viral protein BHRF1, 88% of their designed binders successfully bound when tested in DeepMind's wet lab. Based on the tested targets, AlphaProteo binders hold together roughly 10 times more strongly than those created using existing state-of-the-art design methods, making it a true bioengineering breakthrough. The model is being used for drug development, diagnostics, and biotech applications.

AlphaProteo generating successful binders
Source: Google DeepMind, 2024
Figure 5.1.3

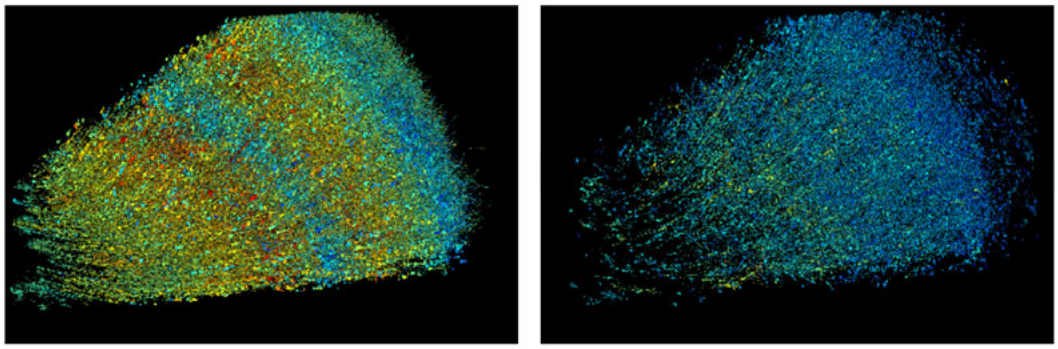


Human Brain Mapping

Synaptically reconstructing a small piece of the human brain

A team at Google's Connectomics project has reconstructed a one-cubic-millimeter section of the human brain at the synaptic level—hailed by Wired as "the most detailed map of brain connections ever made." The sample, taken from an epileptic patient's left anterior temporal lobe during surgery, was imaged with a multibeam scanning electron microscope. Over 5,000 ultra-thin slices (30 nanometers each) captured around 57,000 cells—including neurons, glial cells, and blood vessels—along with 150 million synapses. Figure 5.1.4 visualizes the results: excitatory neurons on the left, inhibitory neurons on the right. To process this massive dataset, the team developed machine learning tools like flood-filling networks (for neuron reconstruction without manual tracing), SegCLR (for cell type identification), and TensorStore (for managing the multidimensional dataset). The dataset is publicly available via Neuroglancer, a web-based exploration tool; and CAVE, a Neuroglancer extension for annotation refinement. This project marks a major step in understanding neural circuitry and could inform future neurological treatments.

3D brain map images
Source: Google Research, 2024
Figure 5.1.4



Virtual AI Lab

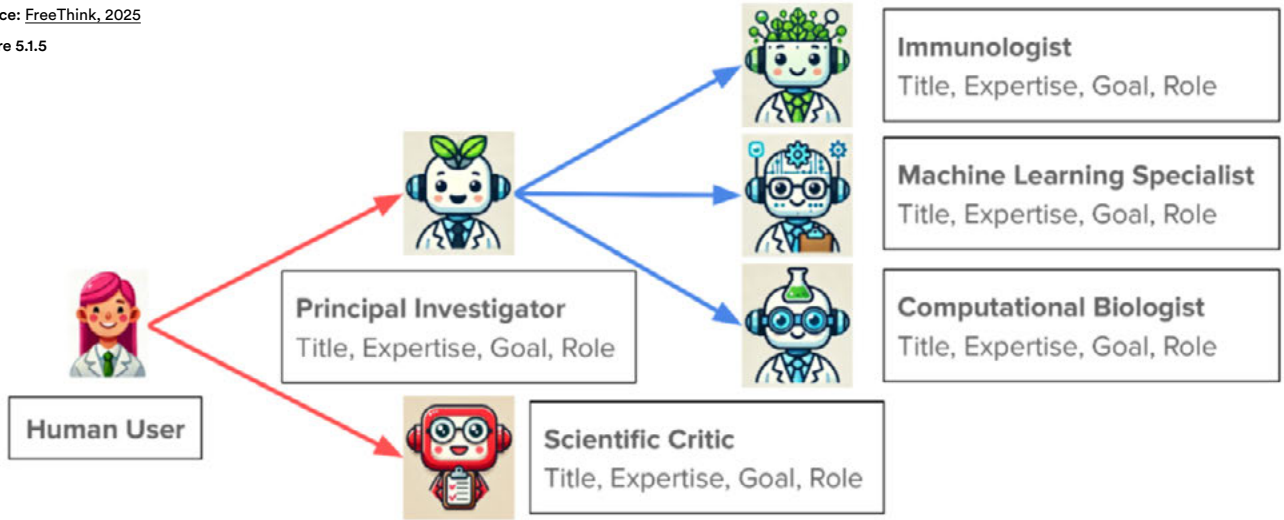
Virtual AI lab supercharges biomedical research

AI's role in science is shifting from a passive tool to an active collaborator. A recent Stanford study introduced a virtual AI laboratory, where multiple AI-powered scientists (technically LLMs) specialize in different disciplines and autonomously collaborate as agents. In one experiment, human researchers tasked this AI-driven lab with designing nanobodies—antibody fragments—capable of binding to SARS-CoV-2, the virus that causes COVID-19. The lab generated 92 nanobodies, with over 90% successfully binding to the virus in validation studies. The virtual lab was structured similar to a computational biology lab, comprising a principal investigator (PI), a scientific critic AI, and three discipline-specific scientists specializing in immunology, computational biology, and machine learning (Figure 5.1.5). The PI model created these expert scientists and guided their research. Tools like AlphaFold and Rosetta were used for protein design, but the real significance of this study lies not in its specific findings, but in demonstrating that an entirely autonomous, LLM-powered lab can generate meaningful scientific discoveries.

Workflow in AI-based lab

Source: FreeThink, 2025

Figure 5.1.5



GluFormer

Continuous glucose monitoring with AI

GluFormer, a foundation model developed by Nvidia Tel Aviv, the Weizmann Institute, and others, analyzes continuous glucose monitoring (CGM) data to predict long-term health outcomes. Trained on over 10 million glucose measurements from nearly 11,000 individuals—most without diabetes—it forecasts health trajectories up to four years in advance. For instance, GluFormer can identify individuals at risk of developing diabetes or worsening glycemic control long before symptoms appear. In a 12-year study of 580 adults, it accurately flagged 66% of new-onset diabetes cases and 69% of cardiovascular-related deaths within their respective top-risk quartiles. The model's results have also generalized across 19 external cohorts (n=6,044) in five countries and diverse health conditions. GluFormer often outperforms standard CGM-based metrics like the glucose management indicator (GMI) (Figure 5.1.6). In the near and long term, models like GluFormer will shift diabetes care from reactive treatment to proactive prevention, enabling earlier clinical intervention.

GluFormer versus glucose management indicator

Source: Lutsker et al., 2024

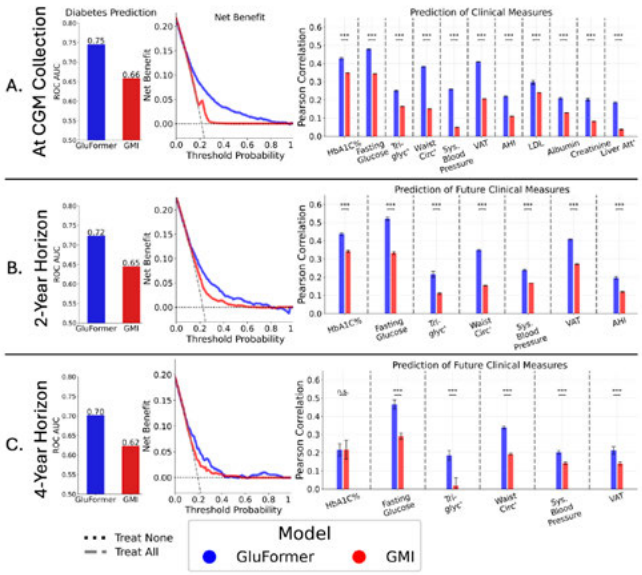


Figure 5.1.6

Evolutionary Scale Modeling v3 (ESM3)

Simulating evolutionary processes to generate novel proteins

EvolutionaryScale's ESM3 is a groundbreaking model designed to generate novel proteins by simulating evolutionary processes. The model was trained on 2.78 billion protein sequences, and hosts 98 billion parameters. Like many other AI models, it is available in three sizes (small, medium, and large) and is available both via API and their partners' platforms. Perhaps ESM3's most notable achievement is designing esmGFP, a new artificial green fluorescent protein which the company estimates would take nature 500 million years to develop. This was done through human-led chain-of-thought prompting. Figure 5.1.7 illustrates the performance of various ESM3 models in generating proteins that satisfy atomic coordination prompts. The results show that larger ESM3 models solve twice as many tasks. ESM3 is also open-sourced, promoting collaboration in synthetic biology and protein engineering projects which hope to use code and data from the project—with applications in drug discovery, materials science, and environmental engineering.

ESM3 models evaluated on protein generation from atomic coordination prompts

Source: ESM3, 2024 | Chart: 2025 AI Index report

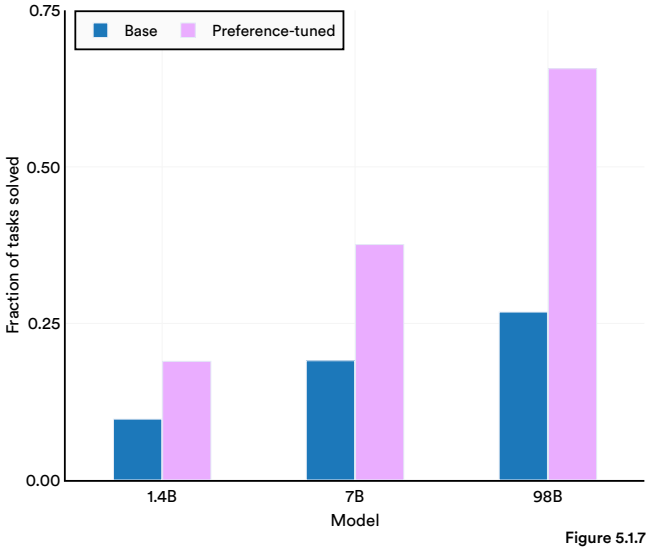


Figure 5.1.7